

SOFTWARE ENGINEERING LAB – 1

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Sem-5, Section J

Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: A resident submits a ticket containing a room number, issue category/severity, and photo attachment; the system creates a unique ticket ID and routes the ticket to the warden queue. Fail: The system accepts a ticket without a room number.	Provides the core mechanism for residents to report hostel maintenance issues.	Specify whether photo attachment is mandatory or optional.
FR-002	Functional	The system shall allow the maintenance warden to view and review submitted maintenance tickets.	High	Pass: When the warden opens the ticket queue, each submitted ticket displays its ticket ID, room number, issue category, severity, and current status. Fail: A submitted ticket is not displayed in the warden queue.	Allows the warden to monitor and manage reported maintenance issues.	Specify whether the warden can filter tickets by severity, category, or status.
FR-003	Functional	The system shall allow the maintenance warden to assign a maintenance ticket to a maintenance staff member.	High	Pass: When the warden selects an unassigned ticket and a maintenance staff member, the system records the staff member as the assignee and displays the updated assignment. Fail: The ticket remains unassigned after the assignment operation.	Ensures that reported issues are dispatched to maintenance personnel.	Specify whether only available maintenance staff can be assigned.
FR-004	Functional	The system shall allow maintenance staff to update the status of an assigned maintenance ticket.	High	Pass: When maintenance staff changes the status of an assigned ticket, the new status is stored and displayed when the ticket is reopened. Fail: The previous status is displayed after the update.	Allows maintenance progress to be tracked.	Specify the allowed ticket statuses.
FR-005	Functional	The system shall allow residents to confirm the resolution of a maintenance ticket and provide a closure sign-off.	High	Pass: When a resolved ticket is opened by its resident, the resident can confirm the resolution and submit a closure sign-off, after which the ticket status becomes Closed. Fail: The ticket becomes Closed without resident confirmation.	Confirms that the reported issue has been resolved and accepted by the resident.	Specify whether residents can reject a resolution and reopen a ticket.
NFR-00	Nonfunctional	The system shall	High	Pass: When a high-priority	Prevents high-	Specify the

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1		trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status change.		ticket remains unchanged beyond 24 hours, an escalation SMS/email is generated. Fail: No escalation notification is generated after the 24-hour SLA is exceeded.	priority issues from remaining unattended beyond the SLA.	recipients and whether both SMS and email are required.
NFR-002	Nonfunctional	The system shall restrict maintenance ticket operations according to the user's assigned role.	High	Pass: Residents can perform resident operations, wardens can review and assign tickets, and maintenance staff can update assigned tickets. Fail: A user can perform an operation outside their assigned role.	Prevents unauthorized access or modification of maintenance operations.	Specify the exact permissions for each role.

Actors

Actor 1 — Hostel Resident

The hostel resident uses the system to:

- Log maintenance tickets
- Provide room and issue information
- Attach photographs
- Specify issue severity
- Confirm issue resolution
- Provide closure sign-off

Actor 2 — Maintenance Warden

The maintenance warden uses the system to:

- View submitted maintenance tickets
- Review reported issues
- Assign maintenance tickets to maintenance staff
- Monitor ticket progress

Actor 3 — Maintenance Staff

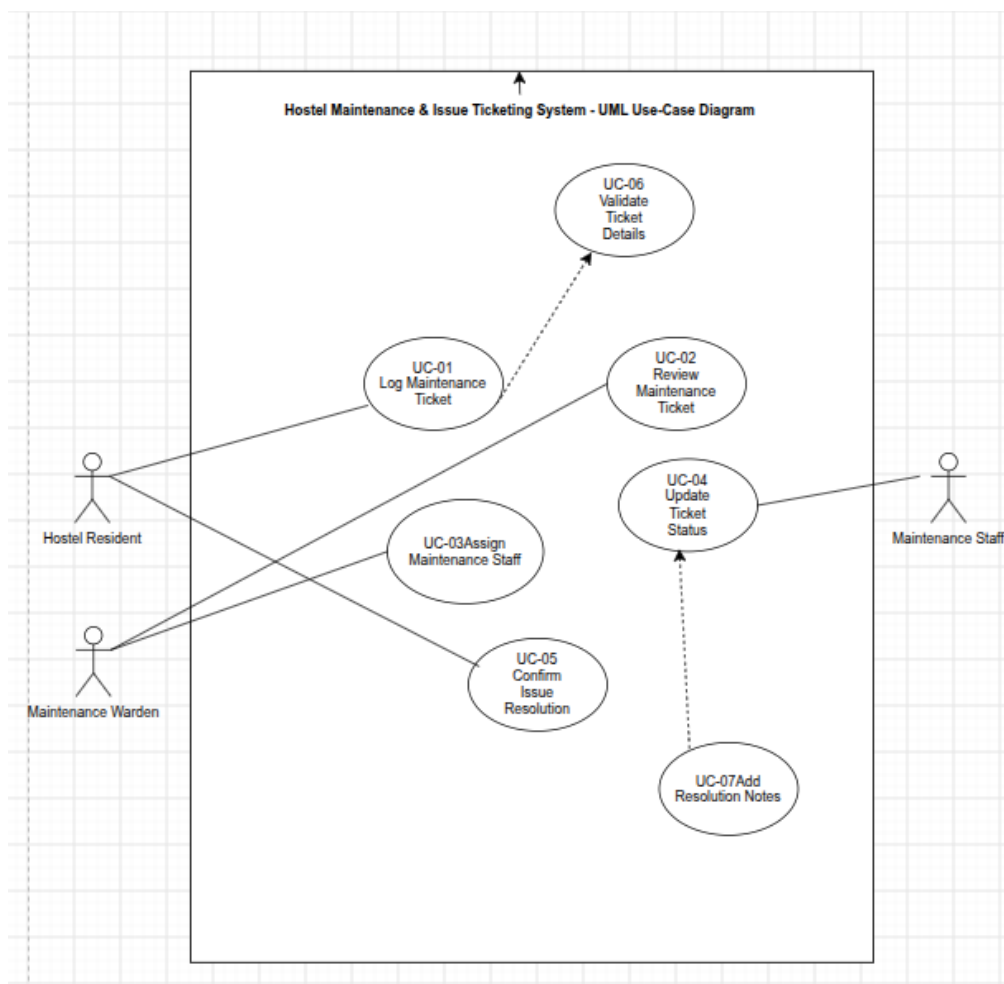
Maintenance staff use the system to:

- View assigned maintenance tickets
- Update the status of assigned tickets
- Provide resolution information

Use Cases

Use Case ID	Use Case	Actor
UC-01	Log Maintenance Ticket	Hostel Resident
UC-02	Review Maintenance Ticket	Maintenance Warden
UC-03	Assign Maintenance Staff	Maintenance Warden
UC-04	Update Ticket Status	Maintenance Staff
UC-05	Confirm Issue Resolution	Hostel Resident
UC-06	Validate Ticket Details	System / included by UC-01
UC-07	Add Resolution Notes	Maintenance Staff / extends UC-04

UML Use Case Diagram



Use-Case flow

UC-01 – Log Maintenance Ticket

Primary Actor: Hostel Resident

Secondary Actor: Maintenance Warden

Preconditions:

1. The resident is authenticated and has access to the Hostel Maintenance & Issue Ticketing System.
2. The resident has a valid hostel room number.
3. The maintenance ticketing service is available.

Main Success Scenario

1. The resident selects "Log Maintenance Ticket" from the system.
2. The system displays the maintenance ticket form.
3. The resident enters the room number and describes the maintenance issue.
4. The resident selects the issue category, such as plumbing, electrical, or carpentry.
5. The resident selects the severity of the issue.
6. The resident attaches a photo of the issue.
7. The resident submits the maintenance ticket.
8. The system validates the entered ticket details.
9. The system creates the maintenance ticket and assigns it a unique ticket ID.
10. The system routes the ticket to the maintenance warden's queue.
11. The system displays a confirmation message containing the ticket ID.
12. The use case ends successfully.

Alternate Flow

1. The system detects that required ticket information is missing or invalid.
2. The system displays an appropriate validation error message.
3. The resident corrects or completes the required information.
4. The resident resubmits the ticket.
5. The system validates the updated information.
6. If the information is valid, the system continues from **Step 9** of the Main Success Scenario.
7. If the information is still invalid, the system displays the validation error again and does not create the ticket.

Post Conditions

1. A unique maintenance ticket ID has been generated.
2. The ticket contains the resident's room number, issue category, severity, and submitted photo.
3. The ticket has been routed to the maintenance warden's queue.
4. The ticket has an initial status of Submitted.